

Fractional Reserve Banking, Unrealizable Capital, and Bank Runs as Temporal Irreversibility

Abstract

Fractional reserve banking is usually analyzed in terms of liquidity, confidence, and systemic risk. This paper proposes an alternative framework: bank runs are not primarily financial or psychological events, but temporal failures. When capital is created through fractional reserves, real assets are acquired and legal states are fixed in time. These operations introduce irreversible historical commitments that cannot be undone by accounting mechanisms. A bank run therefore represents a collective demand for temporal reversal, which is structurally impossible. We define this condition as *temporal irreversibility of financial realization* and show that fractional reserve systems operate as critical systems whose stability depends on the permanent postponement of realization. This framework connects banking instability with the theory of unrealizable capital and the Oniric Value Theorem.

1. Introduction

Modern financial systems are commonly interpreted through monetary aggregates, liquidity ratios, and behavioral confidence models. However, these approaches overlook a fundamental structural property: **temporal irreversibility**.

Once a credit is issued and used to acquire a real asset, the system incorporates a historical event that cannot be undone without coordinated reversal of legal, social, and economic states. This paper argues that fractional reserve banking embeds irreversibility at its core, and that bank runs represent the moment when this irreversibility becomes explicit.

2. Fractional Reserve Banking as a Temporal System

In a full-reserve system, every loan corresponds to pre-existing money. In such a system, rollback is theoretically possible.

In a fractional reserve system:

- Money is created at the moment of lending.
- The asset becomes real and legally fixed.
- The backing never existed in full.

Thus, the system contains **forward-only historical commitments**.

Accounting assumes reversibility.

Reality enforces irreversibility.

This creates a structural temporal mismatch.

3. Capital Realization and Irreversibility

When a mortgage is issued:

1. Ownership of a physical asset changes.
2. Legal registries are updated.
3. Money circulates and disperses.
4. Social and economic expectations adapt.

Reversing this event would require synchronized rollback of independent agents with opposing incentives. Such synchronization is physically and socially impossible.

Therefore, capital created under fractional reserves is **temporally unrealizable in reverse**.

4. Unrealizable Capital

We define *unrealizable capital* as capital whose accounting existence cannot be collectively realized without requiring temporal reversal of real historical events.

This does not mean the capital is fictitious.

It means it is **structurally non-reversible**.

The system remains stable only while realizability remains sparse and unsynchronized.

5. Bank Runs as Temporal Collapse

A bank run is usually described as a loss of confidence.

This paper proposes a different definition:

A bank run is a collective demand for temporal reversal of financial commitments.

Depositors are not only asking for money.

They are asking the system to undo past promises.

The collapse does not occur because money is insufficient.

It occurs because history cannot be undone.

6. Critical Systems Framework

Fractional reserve banking behaves as a critical system:

- Below the realizability threshold, it remains stable.
- Above it, it undergoes phase transition.

The control parameter is not money supply or trust.

It is **realizability density**.

When realizability exceeds system capacity, collapse becomes inevitable.

7. Oniric Value Regime

Before collapse, capital operates in an oniric regime:

- Value circulates.
- Assets appreciate.
- Promises multiply.

But realization is deferred.

The system does not lie.

It dreams.

Collapse is the moment when the dream demands awakening.

8. Temporal Singularity

Collapse occurs when the system reaches a point where future continuity requires backward transitions.

This defines a **temporal singularity** in financial systems.

At this point, no policy intervention can restore reversibility.

Only structural reset is possible.

9. Implications

This framework implies that:

- Financial stability is a function of temporal postponement.
 - Capital accumulation does not require realizability.
 - Systemic crises are temporal, not psychological.
 - Confidence is a narrative substitute for irreversibility.
-

10. Conclusion

Fractional reserve banking does not fail because of mismanagement or fear.

It fails because it violates temporal reversibility.

A bank run is not a loss of trust.

It is the moment when time demands honesty.

11. Final Formulation

Fractional reserve banking is a system of unrealizable capital whose stability depends on the permanent deferral of realization.

Bank runs are not financial panics, but temporal phase transitions triggered by irreversible historical commitments.

Reference Framework

Bresciano, C.

Irreversibility of Experience.

DOI: 10.5281/zenodo.17850757